Supplementary material for "Explanations That Arrive in Time: Deadline-Aware, Control-Loop-Structured Explainable AI for Intrusion Detection in Cyber-Physical Systems"

S1. Pre-specified hypotheses

The following were recorded in `docs/02_research_design.md` before the evaluation runs.

- **H1.** LX achieves higher HR@1 and a lower normalised rank than tag-level attributions aggregated post hoc.
- **H4.** LX beats exact Shapley over data-driven clusters (CLX) and random partitions (RPX) with the same number of groups.
- **H2.** TEA increases HR@k at a fixed delivery time and increases temporal stability.
- **H3.** LX+TEA has deterministic cost, lower latency than sampling explainers at equal or higher accuracy, and the highest DCL and in-time rate.
- **H5a.** Localization degrades less under sensor noise with LX+TEA.
- **H5b.** A decoy captures top-1 less often with LX+TEA.
- **H6.** Random window splits inflate detection metrics relative to chronological, attack-disjoint splits.

A hypothesis is reported as supported only if the Holm-corrected two-sided $p < 0.05$ and the effect has the predicted sign.

Table S1: Validation protocols stated in reviewed XAI-IDS studies that use time-series CPS/IIoT data (as stated in the papers; "–" if not stated in the sections we read). Random proportional splits or k-fold CV without a stated chronological or gapped protocol allow overlapping or adjacent windows on both sides of the split.

Study	Data	Stated protocol
Ha et al. (2022)	gas pipeline	80/20 split
Yazdinejad et al. (2025)	SWaT, gas pipeline, UNSW-NB15	70/30 split
Birihanu and Lendák (2024)	SWaT, HAI, Modbus	80/20 split
Xu et al. (2024)	SWaT, WADI	five-fold CV
Rasheed et al. (2026)	SWaT, WADI, BATADAL, IIoT	5-fold CV, 70/15/15
Panesar et al. (2026)	HIL microgrid scalograms	70/30 split
Shoukat et al. (2025)	N-BaIoT, Edge-IIoTset	70/30 split
Attique et al. (2024)	CICIDS2017, X-IIoTID	70/30 split
Saiyed et al. (2026)	WUSTL-IIoT-2021, CICIoT2023	stratified 5-fold CV
Franco de la Peña et al. (2026)	SWaT	padded segment split (leakage-aware)
Fung et al. (2024)	SWaT, WADI, TEP	normal-only training, attack test sets

S2. Validation protocols in the reviewed corpus

S3. HAI 21.03 loop map

The ten groups used for HAI 21.03 (Section 4) are listed below. Tags constant in training were removed, leaving 57 tags in nine non-empty groups; P1-AUX holds discrete pump-state tags, all constant. The full mapping, with engineering-judgement flags, is in `configs/hai/hai2103_loops.yaml`, and the ground truth is in `configs/hai/hai2103_attacks.csv`.

- P1-PC: pressure;
- P1-LC: return-tank level;
- P1-FC: discharge flow;
- P1-TC: heat-exchanger temperature;
- P1-AUX: pumps;
- P2-SC: turbine speed;
- P2-TC: vibration and trip limits;
- P3-LC: upper-tank level;
- P4-ST: steam-turbine model;
- P4-HT: hydro-turbine model.

S4. Additional results

References

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Table S2: Paired comparisons of LX+TEA against selected configurations (two-sided Wilcoxon signed-rank on (attack, detector) units, Holm correction over all 23 comparisons per row group; $r_{\mathrm{rb}} > 0$ favours LX+TEA). At the alarm TEA has no history, so "LX+TEA" equals LX single-shot.

Data	Metric	Comparator	LX+TEA	Comp.	p_{Holm}	r_{rb}
PCL-Sim	HR@1, alarm	CLX (clusters) + TEA	0.854	0.271	<0.001	+1.00
		Fung ens. + TEA	0.854	0.866	0.101	-0.37
		IG + TEA	0.854	0.856	1.000	-0.20
		KernelSHAP-1024 + TEA	0.854	0.844	0.002	+0.74
		LIME-1024 + TEA	0.854	0.854	1.000	+0.25
		LXsingle-shot	0.854	0.854	1.000	+0.00
		LX-bg + TEA	0.854	0.870	<0.001	-0.77
		PI-CAX + TEA	0.854	0.889	<0.001	-0.49
		RE + TEA	0.854	0.857	1.000	-0.05
		RPX (random) + TEA	0.854	0.542	<0.001	+0.95
PCL-Sim	HR@1, +60 s	CLX (clusters) + TEA	0.823	0.228	<0.001	+0.96
		Fung ens. + TEA	0.823	0.822	1.000	+0.05
		IG + TEA	0.823	0.841	<0.001	-0.69
		KernelSHAP-1024 + TEA	0.823	0.798	<0.001	+0.45
		LIME-1024 + TEA	0.823	0.824	1.000	+0.01
		LXsingle-shot	0.823	0.530	<0.001	+0.83
		LX-bg + TEA	0.823	0.847	<0.001	-0.48
		PI-CAX + TEA	0.823	0.868	<0.001	-0.46
		RE + TEA	0.823	0.805	0.074	+0.20
		RPX (random) + TEA	0.823	0.575	<0.001	+0.92
PCL-Sim	DCL@1(60 s)	CLX (clusters) + TEA	0.516	0.096	<0.001	+1.00
		Fung ens. + TEA	0.516	0.502	<0.001	+0.56
		IG + TEA	0.516	0.518	0.693	-0.33
		KernelSHAP-1024 + TEA	0.516	0.485	<0.001	+0.87
		LIME-1024 + TEA	0.516	0.504	0.001	+0.65
		LXsingle-shot	0.516	0.273	<0.001	+0.98
		LX-bg + TEA	0.516	0.516	1.000	+0.00
		PI-CAX + TEA	0.516	0.515	1.000	+0.02
		RE + TEA	0.516	0.487	<0.001	+0.77
		RPX (random) + TEA	0.516	0.332	<0.001	+0.99
PCL-Sim	In-time@1	CLX (clusters) + TEA	0.481	0.144	<0.001	+1.00
		Fung ens. + TEA	0.481	0.442	<0.001	+0.74
		IG + TEA	0.481	0.499	0.014	-0.66
		KernelSHAP-1024 + TEA	0.481	0.434	<0.001	+0.83
		LIME-1024 + TEA	0.481	0.465	0.022	+0.54
		LXsingle-shot	0.481	0.284	<0.001	+0.91
		LX-bg + TEA	0.481	0.494	0.023	-0.67
		PI-CAX + TEA	0.481	0.490	0.541	-0.08
		RE + TEA	0.481	0.409	<0.001	+0.83
		RPX (random) + TEA	0.481	0.291	<0.001	+0.99
HAI 21.03	HR@1, alarm	CLX (clusters) + TEA	0.790	0.448	<0.001	+0.86
		Fung ens. + TEA	0.790	0.757	0.060	+0.58
		IG + TEA	0.790	0.748	0.052	+0.78
		KernelSHAP-1024 + TEA	0.790	0.738	0.004	+0.87
		LIME-1024 + TEA	0.790	0.755	0.050	+0.81
		LXsingle-shot	0.790	0.790	1.000	+0.00
		LX-bg + TEA	0.790	0.740	0.012	+1.00
		PI-CAX + TEA	0.790	0.930	<0.001	-0.89
		RE + TEA	0.790	0.709	0.060	+0.46
		RPX (random) + TEA	0.790	0.248	<0.001	+1.00
HAI 21.03	HR@1, +60 s	CLX (clusters) + TEA	0.696	0.379	<0.001	+0.84
		Fung ens. + TEA	0.696	0.720	0.600	-0.37
		IG + TEA	0.696	0.724	0.117	-0.61
		KernelSHAP-1024 + TEA	0.696	0.689	0.600	+0.30
		LIME-1024 + TEA	0.696	0.692	0.684	+0.38
		LXsingle-shot	0.696	0.597	0.011	+0.48
		LX-bg + TEA	0.696	0.707	0.691	-0.24
		PI-CAX + TEA	0.696	0.928	<0.001	-0.81
		RE + TEA	0.696	0.608	0.005	+0.51
		RPX (random) + TEA	0.696	0.215	<0.001	+0.95
HAI 21.03	DCL@1(60 s)	CLX (clusters) + TEA	0.528	0.322	<0.001	+0.82
		Fung ens. + TEA	0.528	0.540	1.000	-0.29
		IG + TEA	0.528	0.542	1.000	-0.45
		KernelSHAP-1024 + TEA	0.528	0.512	0.578	+0.45
		LIME-1024 + TEA	0.528	0.522	1.000	+0.33

Table S3: Loop HR@1 per detector family ("at alarm, single-shot / 60 s after the alarm with TEA"; mean over attacks of the per-attack mean over 3 seeds; n: attacks detected by at least one seed).

Data	Detector	n	RE	IG	KernelSHAP-1024	Fung ens.	PI-CAX	LX-bg	LX
HAI	linear	46	0.70 / 0.66	0.85 / 0.78	0.82 / 0.72	0.87 / 0.82	0.93 / 0.93	0.85 / 0.80	0.85 / 0.80
	mlp	45	0.75 / 0.64	0.71 / 0.70	0.70 / 0.70	0.73 / 0.72	0.93 / 0.93	0.71 / 0.68	0.75 / 0.68
	cnn	45	0.73 / 0.60	0.67 / 0.69	0.70 / 0.68	0.70 / 0.70	0.94 / 0.93	0.70 / 0.65	0.75 / 0.67
	gru	45	0.66 / 0.53	0.76 / 0.73	0.74 / 0.65	0.72 / 0.64	0.92 / 0.91	0.70 / 0.70	0.81 / 0.63
PCL-Sim	linear	270	0.89 / 0.80	0.89 / 0.87	0.87 / 0.76	0.88 / 0.82	0.89 / 0.88	0.89 / 0.86	0.89 / 0.84
	mlp	226	0.87 / 0.85	0.87 / 0.86	0.86 / 0.85	0.90 / 0.87	0.94 / 0.88	0.85 / 0.86	0.85 / 0.83
	cnn	253	0.80 / 0.76	0.79 / 0.76	0.78 / 0.78	0.81 / 0.76	0.85 / 0.84	0.83 / 0.79	0.79 / 0.76
	gru	265	0.87 / 0.81	0.88 / 0.87	0.86 / 0.81	0.88 / 0.84	0.89 / 0.87	0.89 / 0.88	0.89 / 0.86

Table S4: Tag-level localization (HR@1 / HR@3, single-shot at the given time after the alarm). H-LX refines the top loop of LX into tags; the other methods rank all tags directly.

Data	Method	+0 s	+30 s	+60 s	+120 s
HAI	RE	0.23 / 0.40	0.13 / 0.27	0.13 / 0.27	0.02 / 0.11
	SM	0.10 / 0.34	0.13 / 0.32	0.13 / 0.26	0.10 / 0.20
	IG	0.28 / 0.41	0.13 / 0.27	0.15 / 0.29	0.07 / 0.15
	KS-1024	0.31 / 0.41	0.15 / 0.29	0.16 / 0.27	0.08 / 0.17
	LIME-1024	0.31 / 0.41	0.15 / 0.28	0.17 / 0.29	0.06 / 0.16
	Fung-ens	0.36 / 0.48	0.23 / 0.37	0.22 / 0.36	0.10 / 0.19
	H-LX	0.31 / 0.49	0.12 / 0.34	0.15 / 0.31	0.06 / 0.18
PCL-Sim	RE	0.38 / 0.76	0.23 / 0.37	0.26 / 0.43	0.26 / 0.42
	SM	0.20 / 0.53	0.18 / 0.37	0.17 / 0.38	0.16 / 0.37
	IG	0.44 / 0.74	0.21 / 0.42	0.25 / 0.47	0.22 / 0.42
	KS-1024	0.44 / 0.73	0.14 / 0.31	0.19 / 0.37	0.18 / 0.33
	LIME-1024	0.44 / 0.73	0.15 / 0.32	0.21 / 0.40	0.20 / 0.36
	Fung-ens	0.28 / 0.75	0.14 / 0.36	0.19 / 0.41	0.18 / 0.38
	H-LX	0.45 / 0.76	0.17 / 0.32	0.21 / 0.39	0.19 / 0.36

Table S5: PCL-Sim loop HR@1 by attack primitive (at alarm, single-shot / +60 s with TEA; n: detected (attack, detector) units).

Primitive	n	RE	IG	LX-bg	LX
CV	135	0.87 / 0.82	0.86 / 0.84	0.87 / 0.82	0.88 / 0.83
CV_PV	160	0.96 / 0.91	0.95 / 0.88	0.95 / 0.88	0.95 / 0.87
CV_ST	159	0.94 / 0.96	0.93 / 0.96	0.94 / 0.98	0.93 / 0.96
PV_BIAS	148	0.93 / 0.85	0.94 / 0.95	0.93 / 0.94	0.93 / 0.93
PV_FREEZE	115	0.56 / 0.54	0.51 / 0.49	0.56 / 0.54	0.50 / 0.46
SP	137	0.86 / 0.82	0.90 / 0.89	0.90 / 0.87	0.90 / 0.86
SP_PV	160	0.79 / 0.66	0.81 / 0.80	0.86 / 0.81	0.81 / 0.76